

KYLE WILSON

Email: kywi4190@colorado.edu

<https://github.com/kywi4190/Projects>

<https://www.linkedin.com/in/kyle-michael-wilson>

TECHNICAL SKILLS

- **Languages:** Python, C, C#, Java, SQL, R, HLSL
- **Frameworks & Libraries:** PyTorch, TorchRL, NumPy, Pandas, ChromaDB, Hugging Face Transformers
- **Tools & Platforms:** Git, Docker, Linux, OpenAI APIs, Unity, Snowflake, Dataiku, Codex

PROJECTS

Financial Document RAG Agent (Python, Agentic AI)

- Built agentic RAG system that ingests SEC 10-K filings; generates cited investment memos with XBRL-grounded accuracy.
- Designed multi-agent pipeline (data, qualitative, synthesis) with LLM tool calling (XBRL lookup) and CRAG self-correction.
- Implemented hybrid retrieval: BM25 + dense vector search via RRF, cross-encoder reranking, numerical query boosting.
- Engineered structure-aware chunking with atomic table preservation; evaluated via RAGAS and financial-domain metrics.

Prompt to Math Visualization Generator (Python, Agentic AI)

- Built a web app that converts natural language prompts into 3Blue1Brown-style narrated, animated math explainer videos.
- Designed multi-agent pipeline (generator, critic, error-handler) with Pydantic-validated schemas as inter-agent contracts.
- Implemented self-healing render loop with typed error classification, targeted LLM repair prompts, graceful degradation.
- Complete automation: LLM code generation, Manim rendering, parallel OpenAI TTS, subtitle sync, and FFmpeg muxing.

Multi-Agent Deeper Research Pipeline (Python, Claude Agent SDK, Agentic AI)

- Built gated multi-agent pipeline mapping research spaces; produces cited decision reports with sensitivity analysis.
- Designed deterministic state-machine orchestrator deploying parallel LLM subagents; uses markdown workspace as state.
- Implemented team of diverse breadth vs depth agents, quality-relevance budgeting, saturation and score-stability stop rules.
- Engineered critic/verifier passes, prosecutor/steelman/frame-check tournament, and a hook-enforced preference quarantine.

Drought Prediction from Beetle Imagery (Python, PyTorch, Foundation Models)

- Built a probabilistic ML system predicting drought conditions (SPEI) from beetle specimen images for an ML Competition.
- Leveraged frozen DINOv3+BioCLIP2 vision foundation models with learned attention pooling and specialized MLP heads.
- Engineered Gaussian NLL loss for joint mean and uncertainty prediction; models output calibrated confidence intervals.
- Automated Optuna Bayesian hyperparameter search across architectures, multi-backbone ensemble and TensorBoard logging

Inverted Double Pendulum Reinforcement Learning (Python, PyTorch, TorchRL, Gymnasium MuJoCo)

- Implemented PPO algorithm to balance double pendulum with continuous control, trained policy over 1M environment steps
- Maintained balance for 1000+ timesteps under deterministic evaluation and verified performance across fresh environments.

EDUCATION

Bachelor of Science in Computer Science, Applied Math, University of Colorado Boulder, Boulder, CO 2023-Present

- Current GPA: 3.92
- Tau Beta Pi – National engineering honor society, invite only for top eighth GPA in Engineering
- **Relevant Courses:** Data Science, Linear Algebra, Algorithms, Software Development Tools, Data Structures
- Sigma Nu Fraternity – Ethical leadership development, personal growth and community service
- Study Abroad: UC3M, Madrid (2025)

Bachelor of Science in Finance, Babson College, Wellesley, MA 2022–2023

- Babson Analyst Initiative (2022) – highly selective equity research and stock pitching program

EXPERIENCE

AI Intern, Prologis, Denver Summer, 2026

- Built standardized Snowflake engineering framework for reliable development, compliance and deployment of Cortex agents
- Developed and tested two text-to-SQL Cortex agents from scratch with robust evaluation suites using the framework's pilot
- Created insights dashboards for enterprise AI credit usage trends through in-depth data wrangling and analysis in Dataiku

Software Development Intern, Alphinat, Montreal Summer, 2024

- Implemented and tested new backend features for a low-code platform using .NET, contributing to production releases.
- Developed and validated CRUD workflows interacting with REST APIs, improving reliability and feature compliance.
- Collaborated in agile sprint planning, gaining experience in professional SWE workflows in a team setting.

ADDITIONAL INFORMATION

- Dual U.S. and Canadian Citizen